

### Exercises, sheet 3.

1. Prove the Ehrenfeucht–Fraïssé theorem.
2. Prove that the following  $L = \{E\}$ -theories have quantifier elimination:
  - $E$  is an equivalence relation with infinitely many equivalence classes of cardinality 7,
  - $E$  is an equivalence relation with 7 infinite equivalence classes.
3. Solve all small exercises about Boolean algebras.
4. Prove that a subset  $X \subseteq S_n(T)$  is clopen if and only if it is of the form  $[\varphi]$  for some formula  $\varphi$  in  $n$  variables. Characterise closed subsets of  $S_n(T)$ . What about  $\text{Stone}(B)$  for a Boolean algebra  $B$ ?
5. Prove that a complete type  $p \in S_n(T)$  is isolated if and only if  $\{p\}$  is a clopen set in  $S_n(T)$ .
6. Use ultraproducts to give another proof of the compactness theorem (this proof generalises easily to continuous logic).
7. Let  $p(\bar{x})$  be a countable collection of formulas in variables  $\bar{x}$  and assume that  $M_i$  are  $L$ -structures such that  $p$  is a type with respect to  $\text{Th}(M_i)$ , for  $i \in \mathbb{N}$ . Let  $\mathcal{U}$  be a non-principal ultrafilter on  $\mathbb{N}$ . Show that  $p$  is realised in  $\prod_{i \in \mathbb{N}} M_i / \mathcal{U}$ .
8. Prove that  $|S_n(A)| \leq 2^{|A|+|L|}$ .

The following is an optional exercise for fans of descriptive set theory. Let  $T$  be a countable theory in a countable language  $L$ . Consider

$$X = \prod_f \omega^{\omega^{n_f}} \times \prod_P 2^{\omega^{m_P}} \times \prod_c \omega,$$

where the products are taken over the sets of function, predicate and constant symbols in  $L$  respectively. Let  $\text{Mod}_\omega(T) \subseteq X$  be the subset consisting of  $L$ -structures on  $\omega$  that are models of  $T$ . Define a topology on  $X$  with basic open subsets of the form

$$\{M \in X : M \models \varphi(\bar{a})\},$$

where  $\varphi(\bar{x})$  varies over  $L$ -formulas and  $\bar{a} \in \omega^{|\bar{x}|}$ . Prove that:

- $X$  is a Polish space.
- $\text{Mod}_\omega(T) \subseteq X$  is a closed subset, hence also a Polish space.
- If a partial type  $p$  is not isolated, then the set  $\{M \in \text{Mod}_\omega(T) : M \text{ omits } p\}$  is a dense  $G_\delta$  in  $\text{Mod}_\omega(T)$ .
- For a countable collection of non-isolated types  $p_1, p_2, p_3, \dots$ , there is a countable model  $M \models T$  that omits all of them.